

The Dirichlet Multiplier Algebra Has Infinite Embedding Dimension

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13 August 2026

Status: literature already answered (full Dirichlet-space subcase).

Original question

In Section 7.5, PDF pages 26–27, of their survey on complete Pick algebras, Salomon and Shalit ask for the minimal $d \in \{1, 2, \dots, \infty\}$ such that an irreducible complete Pick algebra $\mathcal{A}$ is isomorphic to $\mathcal{M}_V$ for some $V \subseteq \mathbb{B}_d$. They emphasize that the answer was unknown even for $\mathcal{A} = \text{Mult}(\mathcal{D})$, the multiplier algebra of the classical Dirichlet space [1].

Explicit answer

Hartz explicitly identifies his result as an answer to the question raised in Salomon–Shalit Section 7.5 [2, p. 2]. His Theorem 4.2, on PDF page 16, states:

There do not exist $d \in \mathbb{N}$ and $V \subseteq \mathbb{B}_d$ such that $\text{Mult}(\mathcal{D})$ is algebraically isomorphic to $\text{Mult}(H_d^2|_V)$.

Since $\mathcal{M}_V = \text{Mult}(H_d^2|_V)$ in the survey’s notation, this excludes every finite embedding dimension under precisely the algebraic-isomorphism notion asked about. The universal complete-Pick representation, also reviewed in both papers, provides an embedding into $\mathbb{B}_\infty$. Consequently the minimal dimension for $\text{Mult}(\mathcal{D})$ is exactly

$$d_{\min}(\text{Mult}(\mathcal{D})) = \infty.$$

The supporting authors knew they were answering the source question; this is therefore an explicit literature answer rather than an agent-inferred reformulation.

Scope

The Dirichlet-space case of the source question is completely resolved. The broader request to determine embedding dimension for arbitrary irreducible complete Pick algebras is not resolved by this identification.

References

- [1] G. Salomon and O. M. Shalit, *The isomorphism problem for complete Pick algebras: a survey*, arXiv:1412.7817 (2014).
- [2] M. Hartz, *Embedding dimension of the Dirichlet space*, New York J. Math. **28** (2022), 154–174; arXiv:2107.12941.